

URI CSC 461 - LocalMAP Replication and Exploration Project

Progress Report

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[GitHub Project Repository Link](#)

Abstract

Dimension reduction algorithms that produce clear and separated clusters that preserve the local structure of a dataset have long been sought after. Dimension reduction algorithms are incredibly relevant to the biomedical sciences and allow domain experts to make new discoveries from high dimensional medical data. Over the course of the last 5 years, many new dimension algorithms have emerged. LocalMAP is a new promising algorithm that aims to preserve the local structure of high dimensional datasets and produce high quality clusters. LocalMAP performed significantly better at dimension reduction, or on par with other models, at typical benchmarking datasets in machine learning. The aim of this report is to replicate a large part of the figures and the experiments in the paper that introduced LocalMAP, as well as apply the algorithm to a new dataset and analyze its performance relative to the same algorithms it originally competed against. The original paper is entitled, "[Dimension Reduction with Locally Adjusted Graphs](#)" by the Duke university professors, Yingfan Wang, Yiyang Sun, Haiyang Huang, and Cynthia Rudin. LocalMAP is compared to the following popular dimension reduction algorithms: t-SNE, UMAP, PaCMAP, LargeVis, TriMAP, PHATE, HNNE, Neg-t-SNE, NCVIS, and InfoNC-t-SNE on many popular benchmarking datasets for machine learning. The primary evaluation metric the paper uses to show LocalMAP outperforming other models is Silhouette score, which demonstrates how well separated clusters are. As well, the paper makes use of SVM score, to further argue how separated the data points are once they have been projected into lower dimensional space. Since the labels of the datasets are known ahead of time, Silhouette Score is used in the paper to measure how well the classes cluster together. We were able to reproduce all of these values, and furthermore provide visualizations for each dataset used. We also provide analysis on the performance of LocalMAP on the MNIST and FMNIST data-sets. Lastly, we evaluated the performance of the LocalMAP algorithm on the Olivetti Faces data-set (used primarily for facial recognition). This dataset is not used by the LocalMAP paper, nor is it a typical dataset used in evaluating models, yet it is very much in line with the other image datasets used in evaluating the models.

In this report, the silhouette scores, SVM scores, and visualizations for each model is replicated on MNIST, FMNIST, 20NG, COIL 20 and USPS. Additionally, LocalMAP is applied to the Olivetti Faces data set, a data set which was not included in the original paper or anywhere else online. This section of the paper compares the performance of LocalMAP against other standard dimension reduction algorithms on this dataset, using silhouette scores, SVM scores, and visualizations as criteria.

Introduction

Background Information

Modern dimension reduction algorithms typically convert datasets into graphs before projecting them onto a lower dimensional space. Edges along the graph denote either similarity or dissimilarity. Attractive forces are applied along edges where two nodes (data-points) are perceived as similar, and repulsive forces are applied where the points appear dissimilar. Points that are perceived to be similar should be drawn together and come closer, and points that are perceived to be different should be drawn apart and separated forcefully. The desired effect of this operation is to cause clusters to form that are well separated. Previous dimension reduction algorithms treat

the graph generated by the algorithm as a ground truth that cannot be altered, whereas LocalMAP aims to reconfigure the weights of the attractive edges, and increase the number of repulsive edges in order to enhance clustering.

LocalMAP map is built on two key insights. The first insight is to remove false positive edges. The presence of many false positive edges will cause an attractive force to be applied to many data points artificially bringing them together and creating clusters that should not exist since the points were misidentified as being neighbors. The second insight that LocalMAP is built on is that as the number of data points in the data set increases, the ratio between the number of NN pairs and FP pairs increases. This means that NN pairs will be over represented. LocalMAP handles this by introducing more FP pairs. The ideal result of introducing more FP pairs are clearer clusters with more distinct boundaries.

Methods and Experiments

The test code and data used to obtain the results of the paper can be found on [the Official LocalMAP GitHub Page](#). Data sets were evaluated using code we wrote in Google Colab. The first stage of testing began by running each model through each data set. In order to get a better estimate of the evaluation metrics, 10 iterations were used to compute the mean silhouette score and mean SVM score. Multiple instances of the same code were run across multiple instances of Colab due to the amount of computing time required to compute all of the embeddings and their associated silhouette scores and SVM scores. TSNE took a significant amount of time, and only includes 3 embeddings are included in the MNIST and FMNIST embeddings due to time constraints. As well, Neg-t-SNE, NCVIS, and InfoNC-t-SNE use 5 iterations instead of 10 on FMNIST and MNIST for a similar reason. These data sets are particularly large and more challenging to compute, so only a smaller portion was considered compared to the other data sets used. The datasets used in our paper will included in the recreation portion of the paper will include the MNIST, FMNIST, 20NG, COIL 20, and USPS. In the second section, the Olivetti Face data set (Eigenfaces dataset) is used, and it will be explained in detail, as it is not relevant to the first section. Computing the different Silhouette Scores and SVM scores was an arduous process. The total amount of time to perform all of the computations, if performed on a single note book, likely would have taken over a full day.

MNIST Dataset

The MNIST dataset is a collection of 70,000 images of hand-written digits from 0 through 9 compiled by the United States Postal service. Each image in the dataset is black and white, and of size 28×28 , and is stored as an array of size $28^2 = 784$ in memory. The resulting size of the dataset is a matrix has $d = 784$ features, and $n = 70,000$ entries.

FMNIST Dataset

The FMNIST dataset is a collection of 70,000 images of grayscale clothing in 10 categories. Each image in the dataset is grayscale, and of size 28×28 , and is stored as an array of size $28^2 = 784$ in memory. The resulting size of the dataset is a matrix has $d = 784$ features, and $n = 70,000$ entries.

COIL 20 Dataset

The COIL 20 dataset is another famous machine learning benchmarking dataset. It is a collection of 20 grayscale objects rotated and placed in different positions. Each object has 72 total images, and images with a 5 degree rotation in between photos. Each image in the dataset is grayscale, and of size 128×128 . In total there are 1,440 images. The dataset therefore has 16384 features corresponding to each pixel, and 1440 entries.

20NG Dataset

The 20NG dataset is a collection of 18,000 news groups on 20 different topics. The data set is an embedding of different news articles. It contains 18,846 samples, and 100 features.

USPS Dataset

The USPS dataset is a collection of 9,298 images of grayscale handwritten digits. It is similar to MNIST. Each image in the dataset is grayscale, and of size 16×16 , and is stored as an array of size $16^2 = 256$ in memory. The resulting size of the dataset is a matrix has $d = 256$ features, and $n = 9,298$ entries.

Other Datasets from the LocalMAP Paper

There are 4 additional datasets in the LocalMAP paper that were unused. Finding proper resources for these datasets, especially ones compatible with Python and data science libraries, were few and far between. Therefore, we were not able to replicate these figures. The datasets all had some medical application, and are **Kang**, **Seruat**, **Human Cortex** and **CBMC**. All of these datasets are large human genomic datasets, each with 1000 dimensions.

Methods

Analytical Plans

Our plan is to use the recreation of the experiments in the original LocalMAP paper to gain a thorough understanding of the capabilities and limitations of this dimension reduction algorithm. We will recreate the Silhouette Scores and SVM scores of the different models. We will then apply this knowledge to the Olivetti Faces (Eigen Faces) dataset, a dataset which generally performs quite well on simpler dimension reduction algorithm. LocalMAP works better when its bias towards separating the clusters is matched by well-separated clusters in the data.

Tools

We will be using python for our codebase. All code will be executed in Google Colab. All of our code is contained in our GitHub repository [GitHub repository Link](#) (already linked at the top too). We will be using Sci Kit Learn for the Silhouette Score metric, matplotlib lib for visualizations of data, and numpy for data manipulation. The following libraries are used to compute the 'phate', 'opentsne', 'umap-learn', 'hnn', 'contrastive-ne', 'trimap', 'tqdm', and 'pacmap'. Some of the code is loosely adopted from the LocalMAP GitHub repository. Switching to the Colab environment essentially eliminated any issues of local installation and getting packages to conform. All notebooks used in this project are only intended to run in colab. They will not work elsewhere.

Machine Learning Methods

This project will use the PaCMAP (includes LocalMap) and sklearn libraries to perform dimension reduction algorithms. These algorithms will create two dimensional representations of multidimensional data which can be visualized and compared using evaluation metrics like Silhouette Score and Support Vector Machine Score. We will also be using the other dimension reduction algorithms from the paper including tSNE, UMAP, HNN, InfoNCE, NegtSNE, NCVi, TriMAP, PHATE and PCA as part of our recreation of some of the paper's tables and figures.

Evaluation Metrics

Silhouette Score and Ground Truth Clusters:

For datasets such as MNIST, ground truth clusters provide a clear and essential metric. MNIST consists of the digits from 0-9, which means that in total 10 digits are presented in the cluster. A good dimension reduction algorithm should thus have 10 clusters when projecting the clusters onto a lower space. Datasets like MNIST and FMNIST datasets have an obvious and expected number of clusters, and a good DR algorithm should preserve the number of ground truth clusters.

Silhouette Score is used in the paper to quantify the quality of clusters following a projection. More mathematically, Silhouette Score quantifies how separated clusters are by calculating the average distance to points in other classes, versus the distance from the point to its own classes. The datasets used in the LocalMAP

paper are labeled.

Each point, a_i in the dataset has its own silhouette score, which is defined by:

$$a_i = \sum_{j \in C_i, j \neq i} \frac{d(i, j)}{|C_i| - 1}$$

Each $d(i, j)$ is the distance between point $i \in C_i$, excluding itself. We consider the average of all of these distances. We next calculate the minimum distance between each point in a class C_i , and a point b_j in one of the other classes:

$$b_i = \min_{k \neq i} \sum_{k \in C_k} \frac{d(i, k)}{|C_k|}$$

The silhouette score of the point i is given by:

$$S_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$

We can then calculate the overall silhouette score of the dataset, which is done by averaging over the N data points in the dataset:

$$S = \frac{\sum_i S_i}{N}$$

If a point has a positive silhouette score, then it follows that $b_i > a_i$, which means that the average distance within the class is less than the lowest distance outside of the class. If points on average have a negative silhouette score, then our score indicates that an algorithm tends to separate points of the same class rather than group them up.

SVM (Support Vector Machine) Scoring:

The process of dimensionality reduction is unsupervised even though the labels exist for the test and training data. In order to judge the overall accuracy of a dimensionality reduction method, the reduced dimensionality data can then be fed into a supervised clustering algorithm. The LocalMAP paper uses a Support Vector Machine (SVM) algorithm on the two dimensional data in order to get an accuracy calculation.

SVM draws lines between two clusters that maximize the distance of all points to that line. For data with multiple classes or data that cannot be separated by only straight lines. The data is extended into a third dimension by non-linear mapping (a kernel) so that hyperplanes can be drawn between each cluster. By offsetting each of the similarly clustered data by a similar amounts and the different clusters each by larger amounts, the separate classes can be grouped. Other than describing the SVM as being optimized globally, almost no explanation is given in the paper for any method or parameters. The 'SVM Score' given for each of the dimension reduction algorithms appears to just be the true positive rate divided by the total number of entries.

For this project we will primarily be using SVM score as the accuracy measure when we are recreating the data from the paper. We will use the SVM modules which are a part of the sklearn package both in our recreations and in any further evaluations of different datasets. In the LocalMAP paper, the problems with SVM as a accuracy measure for dimension reduction algorithms are discussed. It will hide poor dimensionality reduction performance when the dimension reduction algorithm incorrectly splits a cluster into several smaller ones because it will undo that mistake. Additionally there is no measurement of cluster distance, so clusters which are not separated by a poor dimension reduction algorithm may be separated by the SVM. It seems as though the authors felt that this method of accuracy measurement was not properly emphasizing the benefits of LocalMAP.

Preliminary Results

Recreating MNIST Results

We ran the code using the same MNIST set used in the study. Below is a comparison of the figure obtained in the paper versus the figure we obtained:

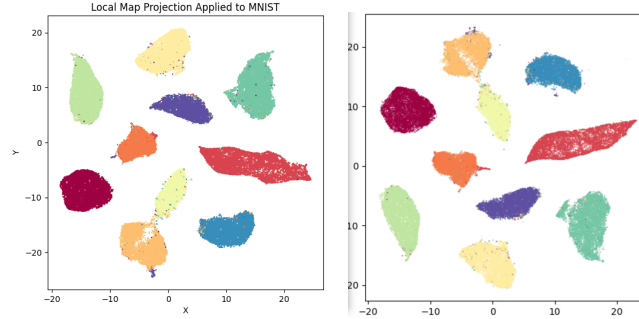


Figure 1: Our recreation (left) of LOCALMAP on MNIST compared to the graph from the paper

The following metrics were reproduced by applying LocalMAP to MNIST. Each column denotes the Silhouette Score and SVM score produced for each embedding. For instance, embedding 1 of LocalMAP on MNIST would correspond to column 1. This calculation of the scores was pretty slow given the problems discussed in the 'Data Pipeline' section of the introduction.

Evaluation Metric	1	2	3	4	5	Average
Silhouette Score:	0.5740	0.5720	0.5752	0.5747	0.5750	0.5742
SVM Score:	0.9744	0.9843	0.9750	0.9743	0.9742	0.9764

Observations on our recreation of LocalMAP applied to the MNIST Dataset:

Our code was able to recreate the structure and separation of the categories present in the example from the LocalMAP paper. The one difference is that the output is flipped vertically and we believe this has something to do with the randomization that is part of the algorithm. This shouldn't matter because the magnitude and sign of the axes is largely irrelevant as long as the clusters are separated properly. In our attempts to run the code locally, numpy gave numeric error warnings due to the sheer size of the embedding sets, however the data produced generally is within the range of the data provided by in the LocalMAP paper.

Recreating FMNIST Results

We ran the code using the same FMNIST set used in the paper. Below is a comparison of the figure obtained in the paper versus the figure we obtained:

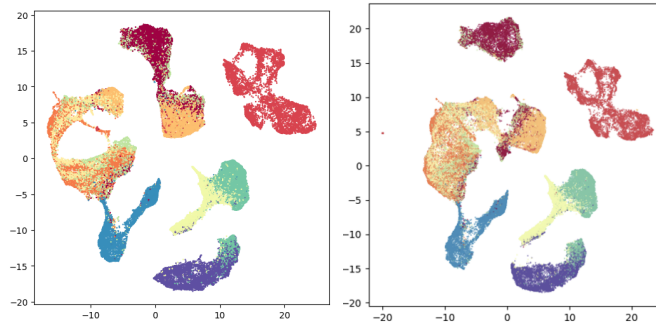


Figure 2: Our recreation (left) of LOCALMAP on FMNIST compared to the graph from the paper

The following metrics were reproduced by applying LocalMAP to FMNIST. Each column denotes the Silhouette Score and SVM score produced for each embedding. For instance, embedding 1 of LocalMAP on FMNIST would correspond to column 1. I would like to note that computing the Silhouette Score and SVM score

Evaluation Metric	1
Silhouette Score:	0.1969
SVM Score:	0.7605

Observations on our recreation of LocalMAP applied to the FMNIST Dataset:

Our code was able to recreate the structure and separation of the categories present in the example from the LocalMAP paper. This output was not flipped vertically compared to the paper unlike the output from other dataset. It is important to note that numpy gave numeric error warnings due to the sheer size of the embedding sets, however the data produced generally is within the range of the data provided by in the LocalMAP paper.

Recreation of the Figures and Tables

Comparing LocalMAP to other algorithms using different types of datasets

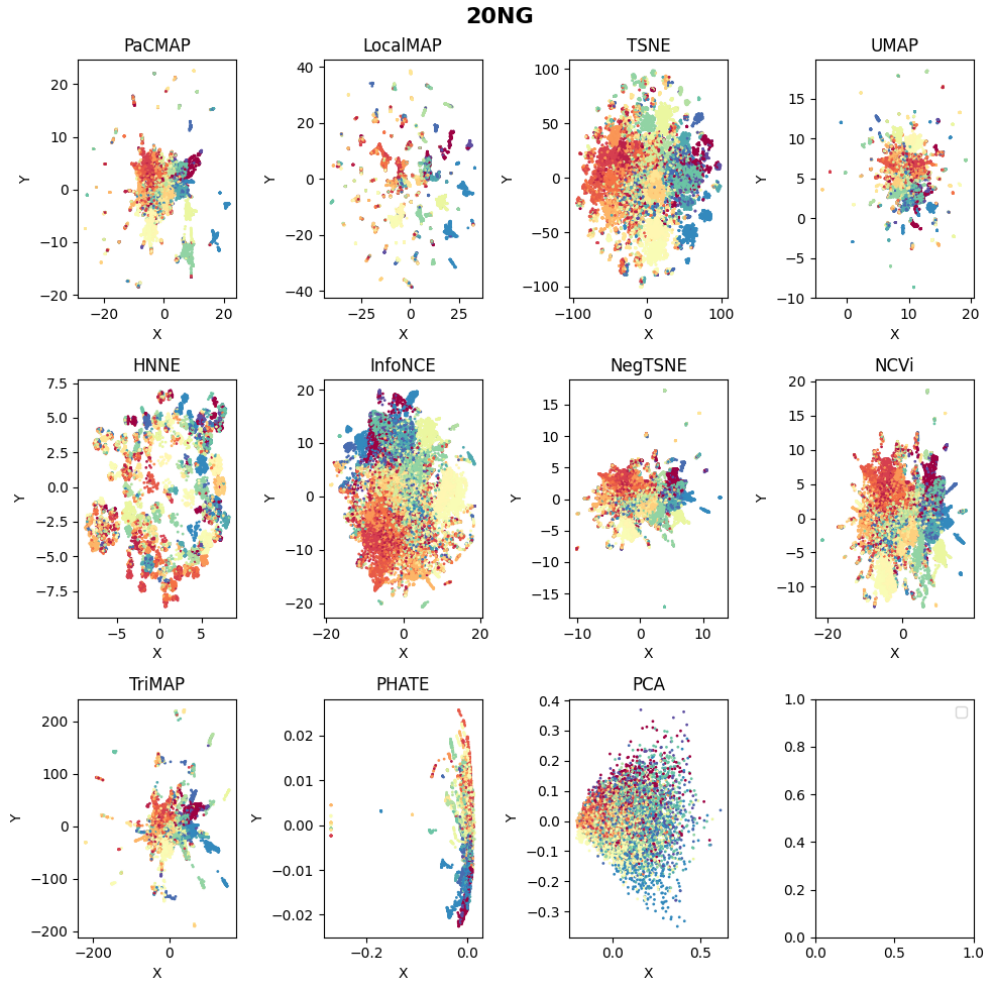


Figure 3: Different dimension reduction algorithms applied to the 20NG dataset with default parameters

The LocalMAP dimension reduction algorithm performed just below the best performing algorithm in terms of accuracy (SVM score) and was the best algorithm in terms of separation (silhouette score). However it did not distinguish itself from many of the other algorithms in particular PaCMAP on which it is based. Although there is more separation in the local map graph that separation doesn't help the algorithm do a better job of accurately separating the newsgroups. While the extra separation introduced by the local map algorithm did not improve the model on the 20NG data, it does show that the model is not distorting datasets that do not fit its assumption.

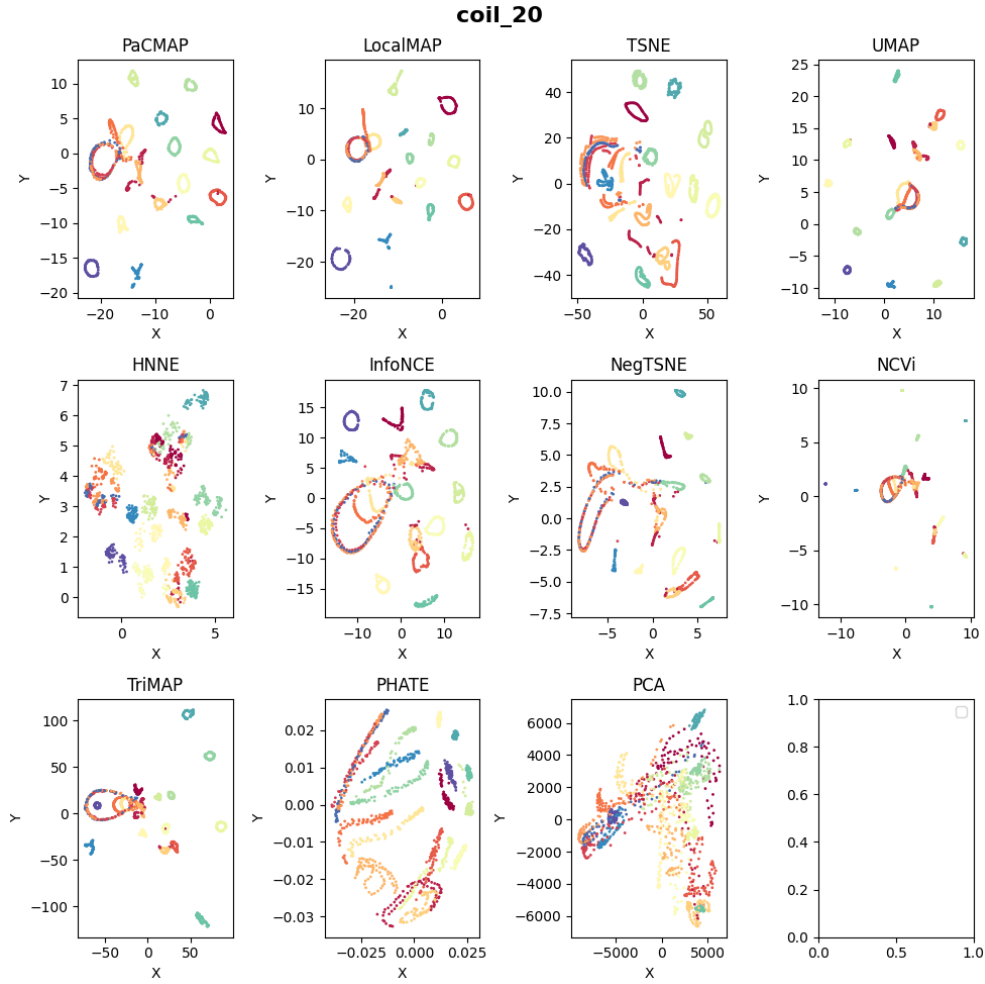


Figure 4: Different dimension reduction algorithms applied to the Coil-20 dataset with default parameters

LocalMAP was among the best performers for this dataset in terms of both accuracy and separation. The separation of features in a picture identification application is a quite good use case for LocalMAP. In particular, this dataset that involves the rotation of the pictures which we believe creates these circular features. The ability of LocalMAP to separate these well could indicate that for other datasets generated a similar way LocalMAP will do a good job of differentiating similar items when rotated.

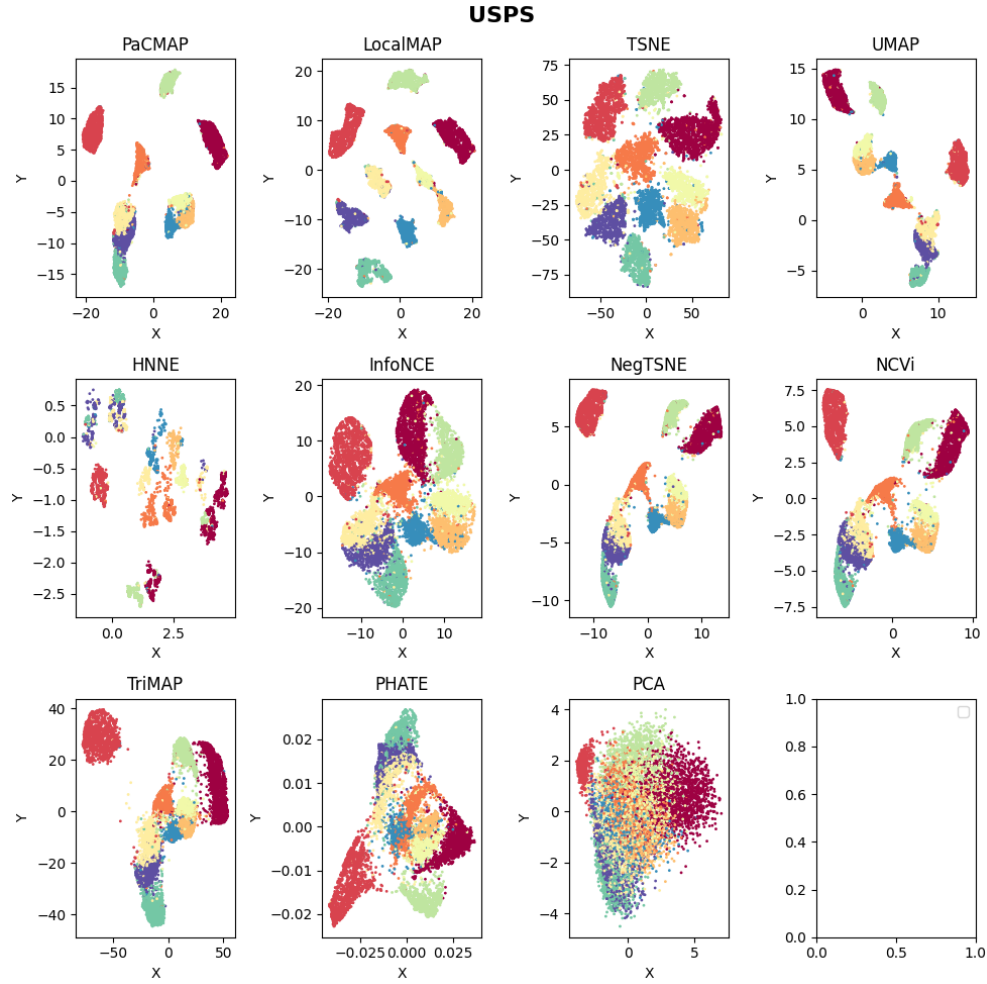


Figure 5: Different dimension reduction algorithms applied to the USPS dataset with default parameters

Well separated data like pictures of handwritten digits continues to be a strength of LocalMAP in particular the silhouette score. Further exploration might include applying this to a database of handwritten letters or other groups of similar features.

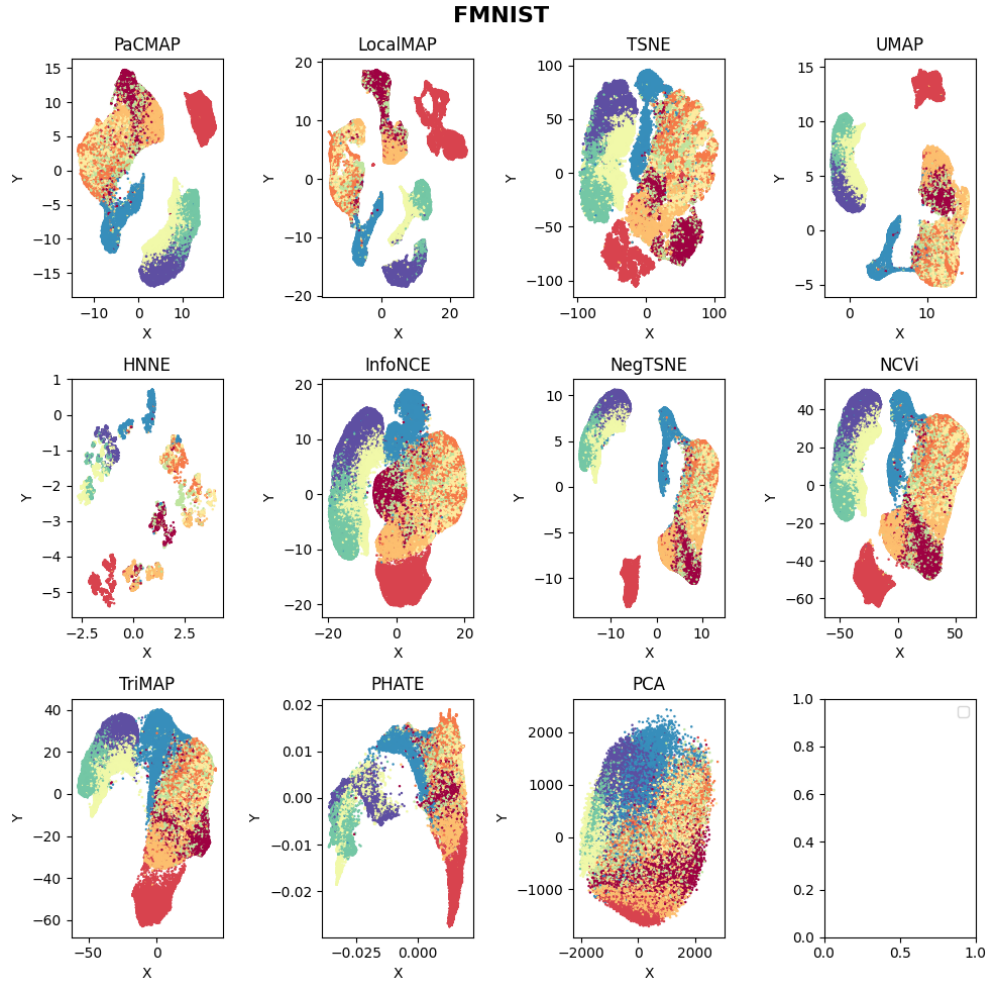


Figure 6: Different dimension reduction algorithms applied to the FMNIST dataset with default parameters

The FMNIST models in this section generally correlate to what is observed in the LocalMAP paper for this appropriate section. This particular visualization took several hours to produce.

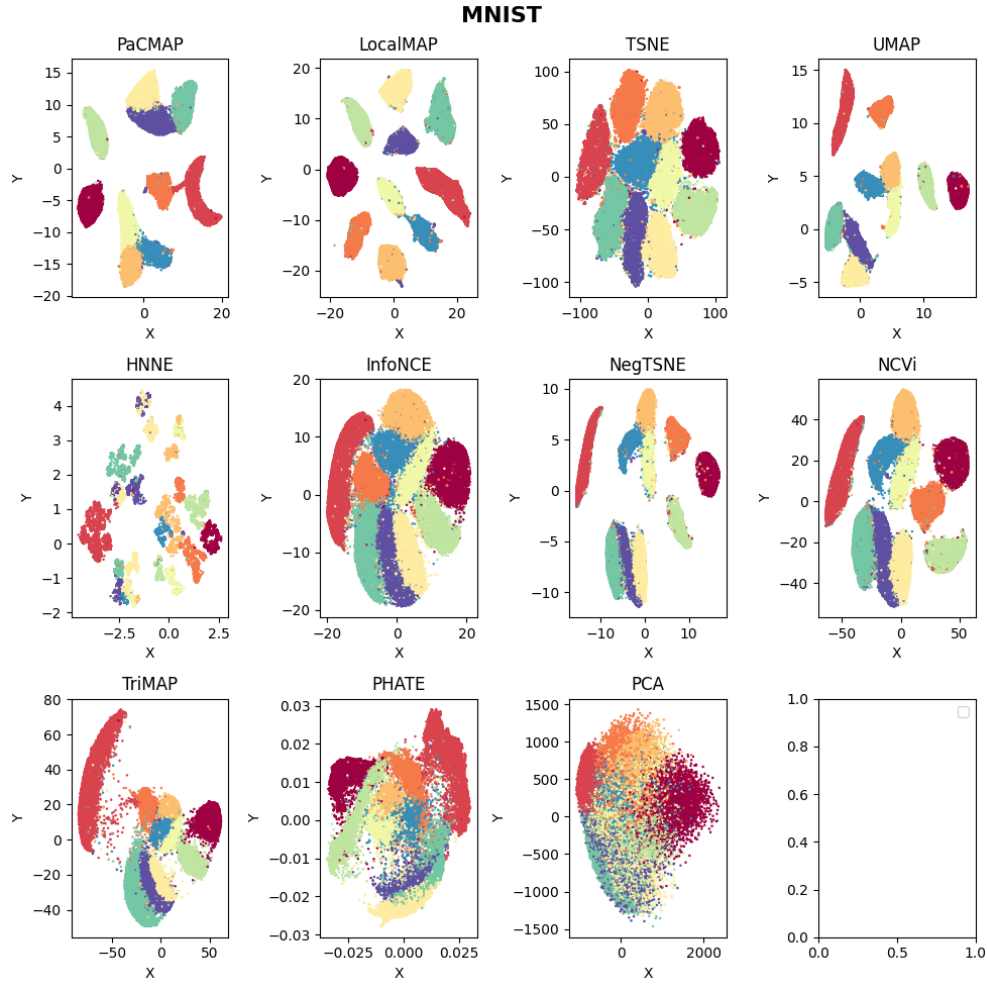


Figure 7: Different dimension reduction algorithms applied to the MNIST dataset with default parameters

We were able to recreate the results for the MNIST dataset to be the same as the paper. The picture clearly shows why LocalMAP would have a excellent silhouette score for this dataset and confirms that PaCMAP and UMAP would have similar SVM scores.

Observations from the Recreated Graphs

We were able to rerun the variety of clustering and dimension reduction algorithms on these datasets and achieve very similar results to the paper. They all looked extremely similar except there is probably something wrong with the FMNIST graphs because we got the same shapes but the colors indicating the groups were extremely scattered. We would have liked to rerun this one with more time and with some clarity into why this problem happened in the first place but we were unable to due to time constraints. These pictures show that LocalMAP is able to create even more separation between clusters that the other dimension reduction algorithms, which is something the more advanced of the other algorithms already seek to do.

Recreated Tables from the LocalMAP Paper

Method	COIL 20	USPS	20NG	FMNIST	MNIST
TSNE	0.42 ± 0.0	0.42 ± 0.0	-0.11 ± 0.0	0.13 ± 0.0	0.38 ± 0.0
LocalMAP	<u>0.55 ± 0.0</u>	0.59 ± 0.0	-0.11 ± 0.0	<u>0.19 ± 0.0</u>	0.57 ± 0.0
NegTSNE	0.44 ± 0.01	0.47 ± 0.0	-0.11 ± 0.0	<u>0.19 ± 0.0</u>	0.47 ± 0.0
PCA	0.01 ± 0.0	0.1 ± 0.0	-0.19 ± 0.0	-0.03 ± 0.0	0.02 ± 0.0
InfoNCE	0.42 ± 0.0	0.37 ± 0.0	-0.12 ± 0.0	0.15 ± 0.0	0.29 ± 0.0
TriMAP	0.47 ± 0.0	0.48 ± 0.0	-0.14 ± 0.0	0.17 ± 0.0	0.41 ± 0.0
NCVi	0.54 ± 0.0	0.45 ± 0.0	-0.11 ± 0.0	0.29 ± 0.0	0.39 ± 0.0
PaCMAP	0.53 ± 0.0	0.56 ± 0.0	-0.11 ± 0.0	<u>0.19 ± 0.0</u>	<u>0.54 ± 0.0</u>
PHATE	0.26 ± 0.01	0.26 ± 0.0	-0.23 ± 0.0	0.1 ± 0.0	<u>0.27 ± 0.0</u>
UMAP	0.57 ± 0.01	0.53 ± 0.0	$-0.14 \pm 0.0 \pm 0.0$	<u>0.19 ± 0.0</u>	<u>0.54 ± 0.0</u>
HNNE	0.24 ± 0.0	0.28 ± 0.0	-0.17 ± 0.0	0.14 ± 0.0	0.11 ± 0.0

Table 1: Silhouette Score for All Models on Different Data Set

(For Silhouette Score) COIL 20 lower bound of 0.56 for silhouette score, upper bound of 0.58. USP lower bound of 0.59, upper bound of 0.69. 20NG, FMNIST, and MNIST, bold the top mean accuracy model, and underline the runner up (due to time constraints).

Method	COIL 20	USPS	20NG	FMNIST	MNIST
TSNE	0.85 ± 0.0	0.96 ± 0.0	0.46 ± 0.0	<u>0.75 ± 0.0</u>	0.97 ± 0.0
LocalMAP	0.83 ± 0.0	0.96 ± 0.0	0.46 ± 0.0	0.76 ± 0.0	0.97 ± 0.0
NegTSNE	0.8 ± 0.0	0.93 ± 0.0	0.43 ± 0.0	0.73 ± 0.0	<u>0.95 ± 0.0</u>
PCA	0.66 ± 0.0	0.56 ± 0.0	0.15 ± 0.0	0.55 ± 0.0	0.47 ± 0.0
InfoNCE	0.8 ± 0.0	0.93 ± 0.0	0.42 ± 0.0	0.73 ± 0.0	0.94 ± 0.0
TriMAP	0.76 ± 0.0	0.94 ± 0.0	0.42 ± 0.0	0.73 ± 0.0	0.96 ± 0.0
NCVi	0.76 ± 0.0	0.93 ± 0.0	0.43 ± 0.0	0.74 ± 0.0	<u>0.95 ± 0.0</u>
PaCMAP	<u>0.84 ± 0.0</u>	0.95 ± 0.0	0.46 ± 0.0	0.74 ± 0.0	0.97 ± 0.0
PHATE	0.83 ± 0.0	0.86 ± 0.0	0.26 ± 0.01	0.65 ± 0.0	0.86 ± 0.0
UMAP	0.81 ± 0.0	0.95 ± 0.0	<u>0.44 ± 0.0</u>	0.74 ± 0.0	0.97 ± 0.0
HNNE	0.78 ± 0.0	0.87 ± 0.0	0.4 ± 0.0	0.68 ± 0.0	0.91 ± 0.0

Table 2: SVM Score for All Models on Data Set

(For SVM) COIL 20 TSNE lower bound of 0.84 and 0.86. USPS TSNE and LocalMAP both have lower and upper bounds of 0.96. 20NG, FMNIST, and MNIST, bold the top mean accuracy model, and underline the runner up (due to time constraints).

Observations and Analysis of the Paper Recreation

Overall, we were able to achieve extremely similar results by recreating the experiments and evaluations of the LocalMAP paper . While we had technical difficulties including long run times and were unable to find versions of the datasets for a couple datasets used in the paper, the comparisons we were able to run came out very similar to the paper. On almost all of the datasets, LocalMAP performed close to or as well as the best algorithms and in particular on several it was significantly ahead particularly in terms of the silhouette score. It is important to note that files were not produced neatly from a single program, but compiled from multiple instances of Collab running in parallel. It is possible that there is some mismatch in the data, which already was found to be the case at least once.

Section II - Olivetti Faces (Eigen Faces)

In this section, the Olivetti Faces dataset is applied to the LocalMAP algorithm. We detail relevant background information about the Eigen faces algorithm, different trials of LocalMAP on the Eigen faces data set, and analysis on the performance of LocalMAP compared to other dimension reduction algorithms on Eigen Faces. The two metrics used in the paper, SVM score and Silhouette Score, will be used to evaluate the quality of the projected data. This section does not aim to recreate a face recognition algorithm, rather it simply observes and evaluates the performance of LocalMAP on an Eigen Faces dataset. The Eigen Faces algorithm performs an initial manipulation of the data from the data set using PCA. Therefore, this section will cover data which has not been transformed according to the Eigen Faces algorithm, and data which has been transformed according to the Eigen Faces algorithm. A surface level explanation of the Eigen Faces algorithm is necessary because facial data is transformed according to a specific procedure that only makes sense within the context of the algorithm.

Eigen Faces Algorithm Overview

Only an overview of the Eigen Faces algorithm will be provided as it is not the focus of this section, but it is necessary to understand the objective and relevance of what is being done. The Eigen Faces algorithm is a PCA based algorithm for facial recognition. Given a dataset of images, $\mathcal{D}_1 = \{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_j\}$, where each $\vec{x}_i \in \mathbb{R}^{n^2}$. Each $\vec{x}_i$ corresponds to a face image, where each is a square and grayscale image of a face presented from various angles. The algorithm generalizes a face from multiple perspectives. Each image is converted into a vector of size n^2 where n is the width and height of the image. There are j many images in the dataset, and each person should appear multiple times so that the algorithm can get a sense of a person's face from different angles. The covariance matrix, C is obtained from the data matrix $\mathcal{D}_1$. The Eigen Faces algorithm constructs the "Face-Space" by taking the first k eigen-vectors of C , where each eigen-vector is denoted by $\vec{v}$:

$$\text{Face Space} = \text{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_j\} = F$$

Each image, $\vec{x}_i$ in C is then approximated by using least squares to solve $F\vec{y} = \vec{x}_i$, where $\vec{y}$ is the encoding of the image with respect to the Face Space. Each face image belonging to a certain individual is averaged in the face space. A new image can be classified by projecting onto the face space, and determining which average face is closest to it.

Motivation for Applying LocalMAP to an Eigen Faces Dataset

The Eigen Faces algorithm is very successful at classifying faces. The algorithm uses PCA to approximate Face Space, the set of all face images, using the span of some eigen vectors. This suggests that the true structure of Face Space must be approximately linear. This is because PCA is a linear algorithm. The face images of the same individual should be roughly distributed in the same area, since they correspond to the same face. We hypothesize, that face images corresponding to the same individual should form a cluster in high dimensional space due to similarity of face images, which implies proximity of the face images. This should be evidenced by the fact that the Eigen Faces algorithm works in the first place. The algorithm is based around the notion that the mean face image of each individual to a new face image. In order for classification to be accurate, it must be the case that face images corresponding to different people must be far enough away from one another such that faces images are not always misclassified. This implies that there is some level of separation between the faces, and that there is almost certainly clustering behavior occurring in the high dimensional space. Therefore, we hypothesize that the Eigen Faces data set should form clusters corresponding to different individuals when Dimension Reduction algorithms are applied to it. Given that clustering behavior is expected, the data set is worthwhile to test LocalMAP on, since the algorithm is meant to excel at forming clusters.

Given that these assumptions about the underlying data are true, it should follow that dimension reduction should produce a cluster for each person involved. If there were 40 people in the training set, there should be roughly 40 clusters to reflect this fact. Some faces may be very similar, and there may be some overlap, or side shots and front shots of the same person should be similar enough that roughly the same amount of clusters is observed in the projected dataset.

The Olivetti Faces Dataset

In order to properly process the Olivetti faces data with LocalMAP, the data needs to be properly prepared. The existing data matrix from SK learn consists of 400 64×64 matrices. Each image was converted into a 1-d array of size $64^2 = 4096$. The resulting data matrix used is of size $64^2 \times 400$, where each pixel of each image is a feature, and there are 400 images considered. There are only 40 unique individuals in the dataset. Each person has 10 photos in the data set. The first ten images correspond to the first person, the second ten images correspond to the second person, the third ten images to the third and so on:

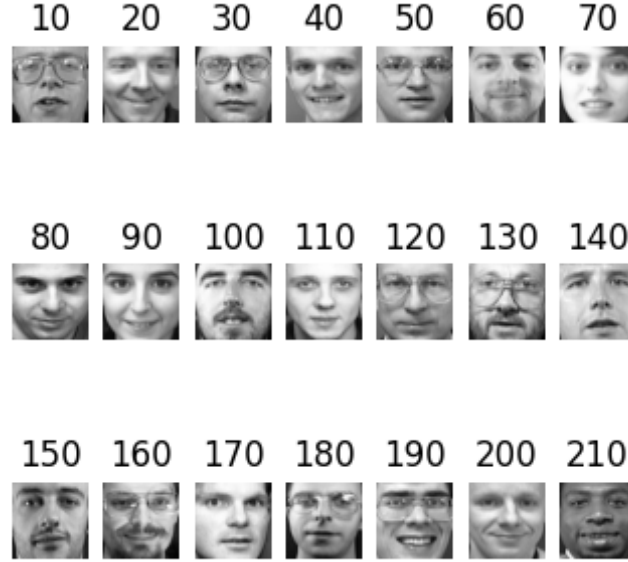


Figure 8: The first 20 different people in the Olivetti Faces data set.

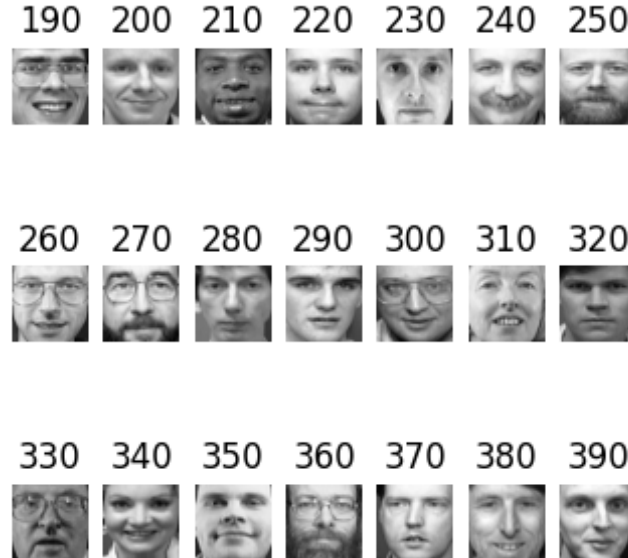


Figure 9: The last 20 different people in the Olivetti Faces data set.

After running through 9 experiments with default parameters, the following embeddings of the Olivetti Faces Dataset were produced:

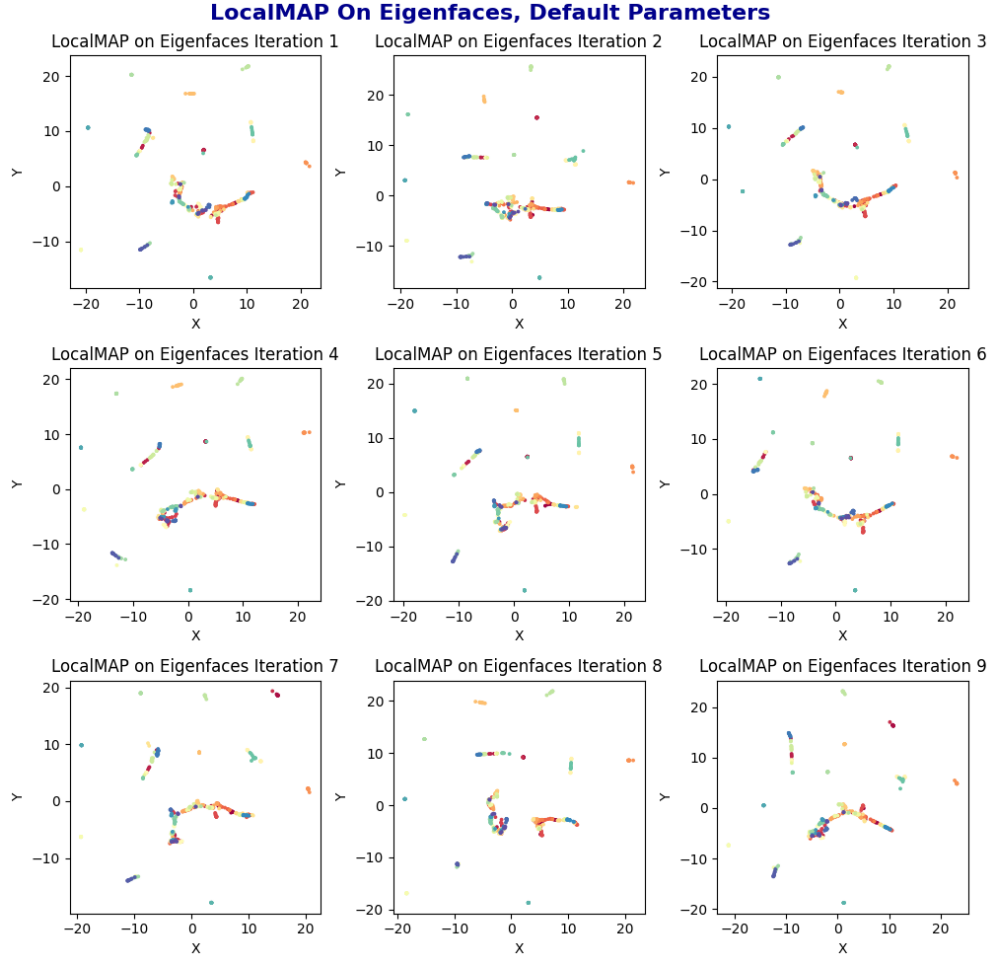


Figure 10: LocalMAP applied to the Eigenfaces dataset using default parameters. Different structures observed are due to randomization used in the LocalMAP algorithm.

To compare the LocalMAP projections with other dimension reduction algorithms, PaCMAP, LocalMAP, TSNE, UMAP, HNNE, InfoNCE, NegTSNE, NCVi, TriMAP, PHATE, and PCA are considered on the unmodified dataset. LargeVIS was left out due to installation issues with the library which were already mentioned at the beginning of the paper.

The associated mean silhouette scores and SVM scores for the dimension reduction algorithms after 10 trials were recorded in the table below. In line with the LocalMAP paper, the model with the highest accuracy is bolded. The model with the highest Silhouette Score evaluation was UMAP, and for a confidence interval of 0.95, it had a lower bound of ≈ 0.118 and an upper bound of ≈ 0.142 . The model with the highest SVM Score evaluation was T-SNE, and for a confidence interval of 0.95, it had a lower bound of ≈ 0.522 and an upper bound of ≈ 0.570 .

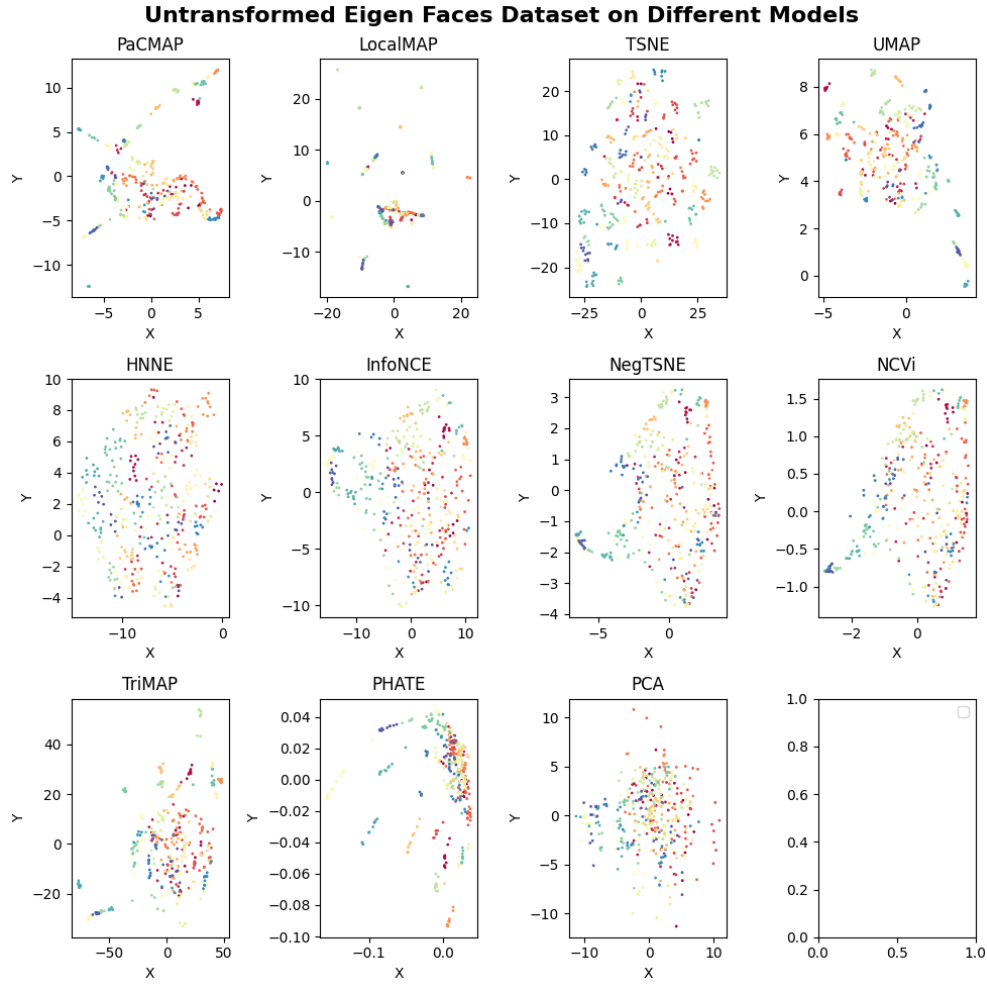


Figure 11: Different dimension reduction algorithms applied to the Eigenfaces dataset using default parameters

Method	Mean Silhouette Score	Mean SVM Score
PaCMAP	0.13 $\pm$ 0.00	<u>0.53 $\pm$ 0.00</u>
LocalMAP	<u>0.12 $\pm$ 0.00</u>	0.52 $\pm$ 0.00
TSNE	0.08 $\pm$ 0.00	0.55 $\pm$ 0.01
UMAP	0.13 $\pm$ 0.00	0.52 $\pm$ 0.01
HNNE	-0.09 $\pm$ 0.00	0.46 $\pm$ 0.00
InfoNCE	-0.08 $\pm$ 0.00	0.40 $\pm$ 0.00
NegTSNE	-0.10 $\pm$ 0.00	0.39 $\pm$ 0.00
NCVi	-0.17 $\pm$ 0.00	0.31 $\pm$ 0.00
TriMAP	0.11 $\pm$ 0.00	0.51 $\pm$ 0.01
PHATE	0.05 $\pm$ 0.00	0.48 $\pm$ 0.00
PCA	-0.26 $\pm$ 0.00	0.21 $\pm$ 0.0

Table 3: Silhouette and SVM scores for different dimensionality reduction methods.

Discussion of LocalMAP Performance on Unmodified Eigenfaces Dataset

All of the different Dimension Reduction models had difficulty separating the dataset. The assumption that the data may be cleanly separable and partitioned neatly into 40 clusters in higher dimensional space may be incorrect or derive from the fact that the data during this stage of experiments was not transformed according to the initial steps of the Eigen Faces algorithm. The current models did not achieve a Silhouette Score higher than 0.13. The model which best performed on the unmodified Eigenfaces set was UMAP, with a mean silhouette score

of 0.13. LocalMAP’s silhouette score was not statistically different than the best models on the dataset. For SVM score, TSNE had the highest accuracy of 0.55, and PacMAP had an accuracy that was not statistically different. LocalMAP did not have a particularly high mean SVM score compared to other models, but still achieved a higher accuracy than many others, and at least seems to be competitive.

In the replication portion of this study, a high silhouette score and SVM score correlated to a high quality visual cluster. In this section, LocalMAP’s relatively high mean silhouette score does not appear to reflect its quality of clustering. LocalMAP seems to cluster quite poorly compared to other models. LocalMAP forms a central cluster consisting of the most of the face images, and very small tight nit clusters of a few face images away from the central cluster. Other dimension reduction models are significantly more diffuse, but appear to do a better job of keeping points separated from one another and closer to their own kind. A key component of the LocalMAP algorithm is adding additional FP pairs, which adds additional edges in the graph of the dataset and applies repulsive forces along them. This should cause clusters to be visually separated from one another. The many points surrounding the central cluster are likely the result of adding additional FP pairs, which cause points to be artificially further from each other. The images on the outside are likely outliers in some way, and taken from a different angle, or the individual might be doing some face in these images that differs from the others.

None of the dimension reduction algorithms produce a clear visualization that reflects an expected 40 ground truth clusters corresponding to each unique individual in the data set. LocalMAP destroys any meaningful structure that can be obtained from observing the visualization, whereas the other two top algorithms, UMAP and PacMAP, keep data relatively well separated. TSNE seems to do a good job at visually handling the data. While clusters aren’t really formed in any of these visualizations, TSNE maintains that face images corresponding to the same individual correspond to at least the same regions. As well, in UMAP and PaCMAP, there are many regions that have face images from many different face classes, suggesting poor separation. In TSNE’s visualization, it appears that a single face image produces multiple clusters that are well separated. This may correspond to a face image from one person being taken from different angles, and thus these face images would tend to coalesce in the same area. Nonetheless, TSNE visually produces better clusters than LocalMAP since it maintains some structure of how faces are distributed. LocalMAP doesn’t care about the structure of how the faces are distributed, and condenses them into the same central mass. Portions of the UMAP visualization also seem to do a good job at keeping data, but run into similar problems of having too many face images from different classes jumbled in a region.

LocalMAP Using Transformed Eigenfaces Data

For the second phase of the experiment, we hypothesize that we consider transformed data. The poor performance of all of the DR algorithms in the first part of the experiment may be improved by performing the initial steps of data manipulation used in the Eigen Faces algorithm. Instead of directly projecting the raw data, $\mathcal{D}_1$ down into two dimensions using a DR algorithm, we first perform PCA on the dataset, and form the Face Space. The Face Space is a subspace, and any face image should ideally live in this subspace. Every face image in the Dataset is projected onto the Face Space, yielding a new dataset, $\mathcal{D}_2$. Each DR algorithm performs dimension reduction on $\mathcal{D}_2$, as opposed to $\mathcal{D}_1$ as in the first part of this experiment.

Let $\mathcal{D}_1 = \{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_j\}$, where each $\vec{x}_i \in \mathbb{R}^{n^2}$ is an original face image. Let $\mathcal{D}_2 = \{\vec{y}_1, \vec{y}_2, \dots, \vec{y}_j\}$, where each $\vec{y}_i \in \mathbb{R}^{n^2}$ is a transformed face image corresponding to the original image $\vec{x}_i$. $\mathcal{D}_2$ can be obtained using the following procedure. Let C be the covariance matrix obtained through $\mathcal{D}_1$ by using PCA. Since $C = C^T$, it follows that $C = X\Lambda X^T$ where X is orthogonal, and Λ is the diagonal matrix of principal values. We select the first k eigen-vectors from X , and have a basis for Face Space. The number of vectors chosen for Face Space betters the approximation, and the amount can vary. This is a parameter that needs to be taken account of when investigating the performance of the DR algorithms. Any face image, $\vec{x}_i$, can be approximated as a linear combination of the eigen-vectors from Face Space, which thus gives $\vec{y}_i$. Therefore, we have obtained $\mathcal{D}_2$.

The varying number of eigen-vectors can be handled by considering 5 new eigen-vectors at a time. Each iteration, 5 eigen-vectors are added to the basis. Computing a visualization of each dataset for each new eigen-vector would take too much time to process for the 11 different models, so only consider up to 100 eigen-vectors is considered. Eigen vectors beyond 100 are not considered, since at $k = 100$ the projections already appear to significantly

match the behavior of the original data set. Adding more vectors beyond this point would be redundant. The resulting visualization ends up looking extremely similar to the unmodified data when considering 100 eigen-vectors, and therefore searching beyond that would be redundant. The following visualizations were produced with $k = 10, k = 50$, and $k = 100$ eigen vectors (other visualizations are included on the GitHub Repository for this paper, but they are not particularly insightful):

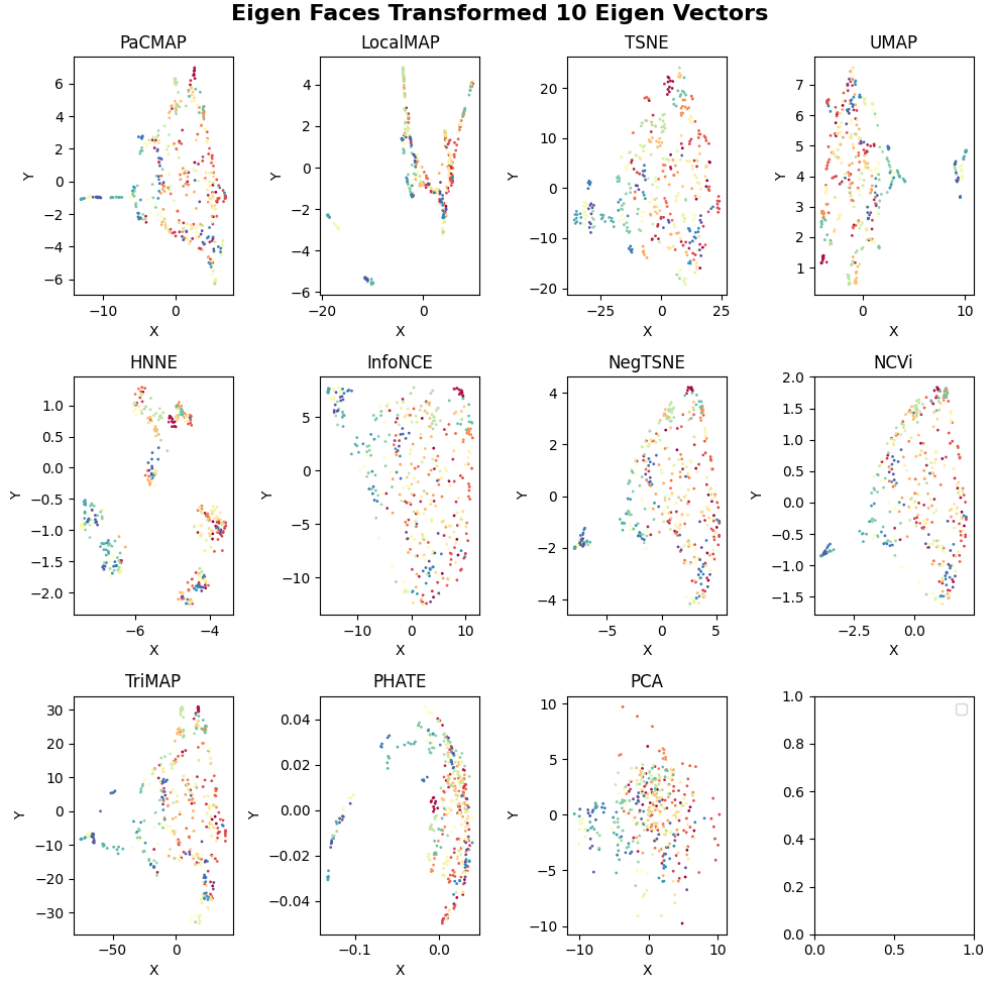


Figure 12: LocalMAP and Other Models on Transformed Eigen Faces set with 10 Eigen Vectors

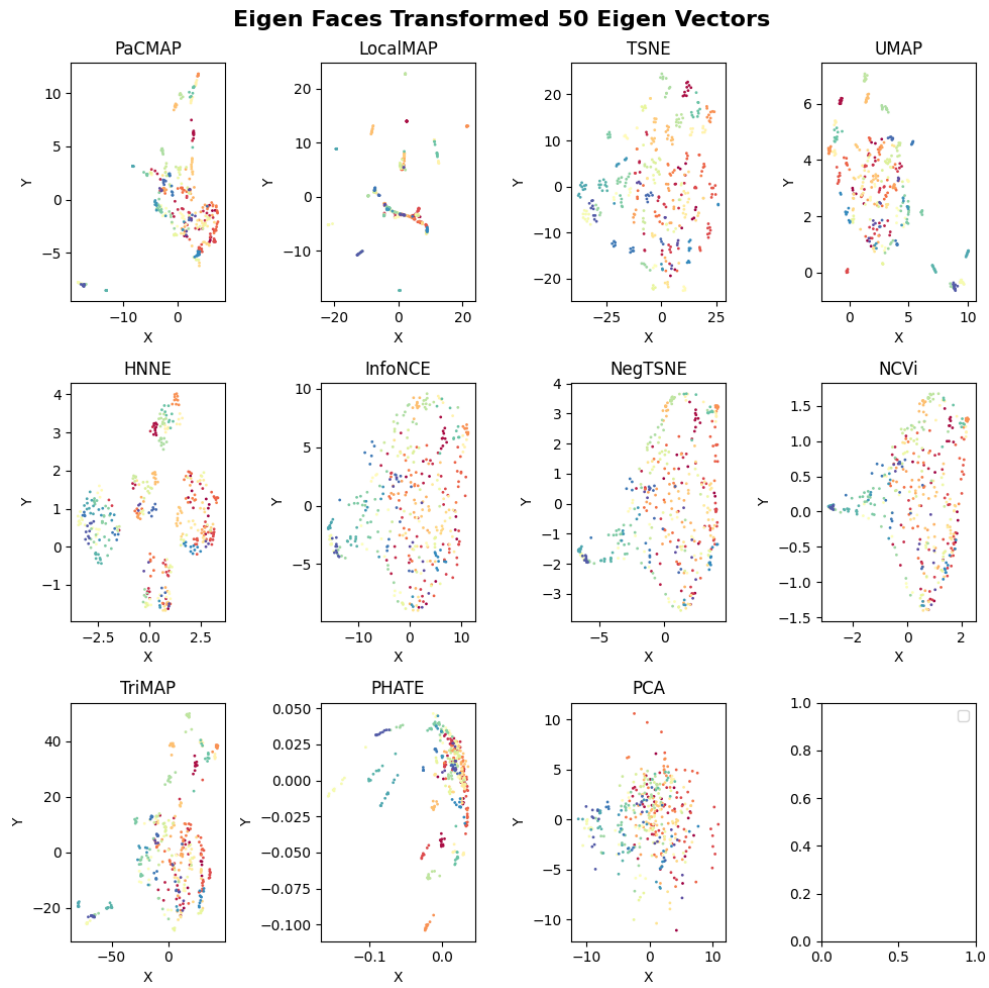


Figure 13: LocalMAP and Other Models on Transformed Eigen Faces set with 50 Eigen Vectors

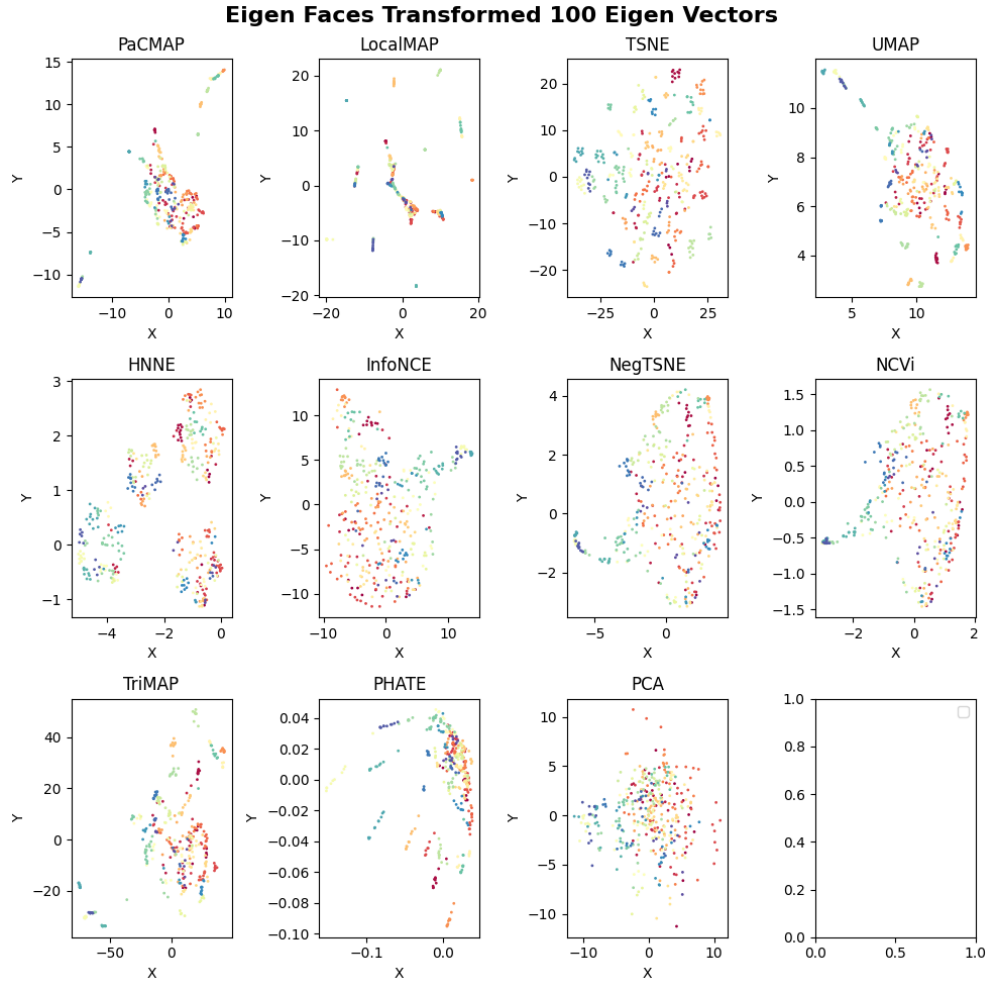


Figure 14: LocalMAP and Other Models on Transformed Eigen Faces set with 100 Eigen Vectors

The following are the associated Silhouette Scores and SVM scores for all of the different trials performed:

Method	k=10	k=50	k=100
TSNE	-0.13 $\pm$ 0.0	0.07 $\pm$ 0.01	0.07 $\pm$ 0.01
LocalMAP	-0.17 $\pm$ 0.01	0.07 $\pm$ 0.0	0.09 $\pm$ 0.01
NegTSNE	-0.22 $\pm$ 0.0	-0.1 $\pm$ 0.0	-0.1 $\pm$ 0.0
PCA	-0.25 $\pm$ 0.0	-0.26 $\pm$ 0.0	-0.26 $\pm$ 0.0
InfoNCE	-0.19 $\pm$ 0.0	-0.08 $\pm$ 0.0	-0.08 $\pm$ 0.0
TriMAP	-0.15 $\pm$ 0.0	0.06 $\pm$ 0.0	0.08 $\pm$ 0.0
NCVi	-0.24 $\pm$ 0.0	-0.17 $\pm$ 0.0	-0.17 $\pm$ 0.0
PaCMAP	-0.13 $\pm$ 0.0	0.09 $\pm$ 0.0	0.11 $\pm$ 0.0
PHATE	-0.22 $\pm$ 0.0	0.01 $\pm$ 0.0	0.04 $\pm$ 0.0
UMAP	-0.1 $\pm$ 0.0	0.1 $\pm$ 0.0	0.13 $\pm$ 0.0
HNNE	-0.34 $\pm$ 0.0	-0.05 $\pm$ 0.0	-0.11 $\pm$ 0.0

Table 4: Silhouette Scores for $k = 10, k = 50, k = 100$ many eigen vectors

Method	k=10	k=50	k=100
TSNE	0.37 $\pm$ 0.0	0.53 $\pm$ 0.01	0.56 $\pm$ 0.0
LocalMAP	0.3 $\pm$ 0.0	0.47 $\pm$ 0.01	0.5 $\pm$ 0.01
NegTSNE	0.22 $\pm$ 0.0	0.37 $\pm$ 0.0	0.39 $\pm$ 0.0
PCA	0.22 $\pm$ 0.0	0.22 $\pm$ 0.0	0.21 $\pm$ 0.0
InfoNCE	0.26 $\pm$ 0.0	0.41 $\pm$ 0.0	0.43 $\pm$ 0.0
TriMAP	0.32 $\pm$ 0.0	0.46 $\pm$ 0.0	0.48 $\pm$ 0.01
NCVi	0.23 $\pm$ 0.0	0.29 $\pm$ 0.0	0.31 $\pm$ 0.0
PaCMAP	0.34 $\pm$ 0.0	0.5 $\pm$ 0.01	0.51 $\pm$ 0.01
PHATE	0.26 $\pm$ 0.0	0.45 $\pm$ 0.0	0.49 $\pm$ 0.0
UMAP	0.34 $\pm$ 0.01	0.47 $\pm$ 0.01	0.51 $\pm$ 0.01
HNNE	0.27 $\pm$ 0.0	0.42 $\pm$ 0.0	0.44 $\pm$ 0.0

Table 5: SVM Scores for $k = 10, k = 50, k = 100$ many eigen vectors

Discussion on Performance of LocalMAP on Transformed Data

The highest mean score model is bolded, and all models which are not statistically different than this model are underlined. If there are no other models that are not statistically different, the model with the second highest accuracy is underlined. The highest mean Silhouette Score model was UMAP for $k = 10, k = 50$, and $k = 100$. When calculating confidence intervals, a confidence interval of $p = 0.95$ was used. For $k = 10$, UMAP achieved a score of 0.13, with a lower bound of -0.11 , and an upper bound of -0.09 for its confidence interval. No model falls within these boundaries, so the second best models are TSNE and PaCMAP tied at -0.13 . For $k = 50$, UMAP obtained 0.10 mean Silhouette Score, with a lower bound of 0.09 and upper bound of 0.11. For $k = 100$, UMAP achieved a mean silhouette score of 0.13 with lower bound 0.12 and upper bound 0.14.

TSNE was the top performing model for SVM score across all iterations. Overall, performing transformations to the data did not seem to stop UMAP or TSNE from being the top performing models across both iterations. Performance on $k = 10$ for all datasets was particularly poor, due to the fact that there were not enough basis vectors in the Face Space to adequately represent each face image. As k grew to 50 and 100, there would be more than enough. Visually, the projections appear to match that of the unmodified sets, and tend to have the same behaviors. The scores all seem to approach that of the results of the original. The data for this section is also strictly linear. Performing PCA on the data matrix did not seem to better any of silhouette scores or SVM scores. All of the scores in this section are considerably lower than that of the unmodified set.

LocalMAP still performs relatively well at the Silhouette score compared to other models. The way LocalMAP tends to handle data is for $k = 50$ and $k = 100$ is very similar to that of the unmodified data. As the number of eigen vectors increases the behavior of the modified dataset approaches that of the unmodified one. The projections are therefore quite similar and did not seem to improve much. TSNE still does a better job at keeping data separated and maintaining some semblance of clustering, even if it partitioned items from the same dataset into multiple groups. This is more insightful than LocalMAP which condenses all of the data into a single strip across both iterations. HNNE also exhibit clustering behavior, whereas when unmodified data was used HNNE created a diffuse mess. The clusters consist of different points, so the model still overall does a poor job at clustering.

Summary of LocalMAP’s Performance on Eigenfaces

LocalMAP performed well in silhouette score across both sections of the experiment. LocalMAP performed better on the untransformed data compared to the transformed data. On the untransformed data LocalMAP was not statistically different than the highest accuracy model. On the transformed data, LocalMAP was not statistically different than the highest accuracy model for $k = 50$ and $k = 100$. LocalMAP achieved a higher accuracy than some of the other models on SVM score, but did not ever achieve the highest SVM score and never achieved an SVM score which was not statistically different than the highest model. The quality of visual clustering is subjective, and Silhouette Score is used to provide an objective metric to dimension reductions. However, despite the fact that LocalMAP performed relatively well in Silhouette Score, its clusters seemed to be of particularly poor visual quality in these experiments. Across all iterations, LocalMAP tended to cluster all of the data into a main central region, and disperse some points around that main area. Little can be inferred about the structure of face images when all of the data is compressed into the center. While other models may produce clusters that are more meaningful, and better preserve patterns and information such as TSNE, silhouette score does not

appear to reflect this fact. LocalMAP excels at optimizing silhouette score, however it does not seem to contribute to overall better visual clustering here.

Summary

LocalMAP is a relatively new algorithm dimension reduction algorithm that's purpose is to achieve greater accuracy and separation of clusters by emphasizing the importance of in-cluster distances (assumed True Positives) over between-cluster distances (assumed False Positives). It is based on the already quite advanced PacMAP algorithm but expands it to create greater separation between clusters and in particular maximize silhouette score. Distinct clusters that preserve local structures are a goal of any advanced dimension reduction algorithm and LocalMAP takes this a step further to tune its optimization function towards maximizing these distances.

We were able to recreate most of the graphs and tables from the paper using the code provided on the paper's GitHub and we achieved almost the same results. We did have a small trouble with the FMNIST dataset and with some dependencies and some very long run times on our un-optimized hardware. Overall our exploration of recreating these findings taught us a lot about how the paper implemented their evaluations and experiments and there were many ways in which the paper intelligently coded their experiments to enable flexible operation of the code.

Our new application of the LocalMAP algorithm to the Olivetti Eignen faces dataset provided us with some new but also parallel insights on the strengths, shortcomings and overall performance of the LocalMAP algorithm. The focus on cluster separation in the LocalMAP algorithm did not do as good a job as t-SNE, PacMAP or UMAP across the different levels of transformation although its performance was decent on this algorithm, often near the top and quite ahead of the other simpler algorithms and some of the more complicated dimension reduction algorithms. The focus on separation and the minimization of assumed false positive connections probably contributed to an enhanced false positive rate although not enough to render the method completely unusable. There was little difference between LocalMAP's performance with different amounts of transformation by the number of eigenvectors. Human faces are more similar to each other than the best performing datasets from the initial paper like the handwritten numbers datasets and finding datasets that exhibit similar characteristics to those datasets will probably be the best use of LocalMAP.

References

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- [2] Turk, M., Pentland, A. *Eigenfaces for recognition*. Journal of Cognitive Neuroscience, Jan. 1st, 1991